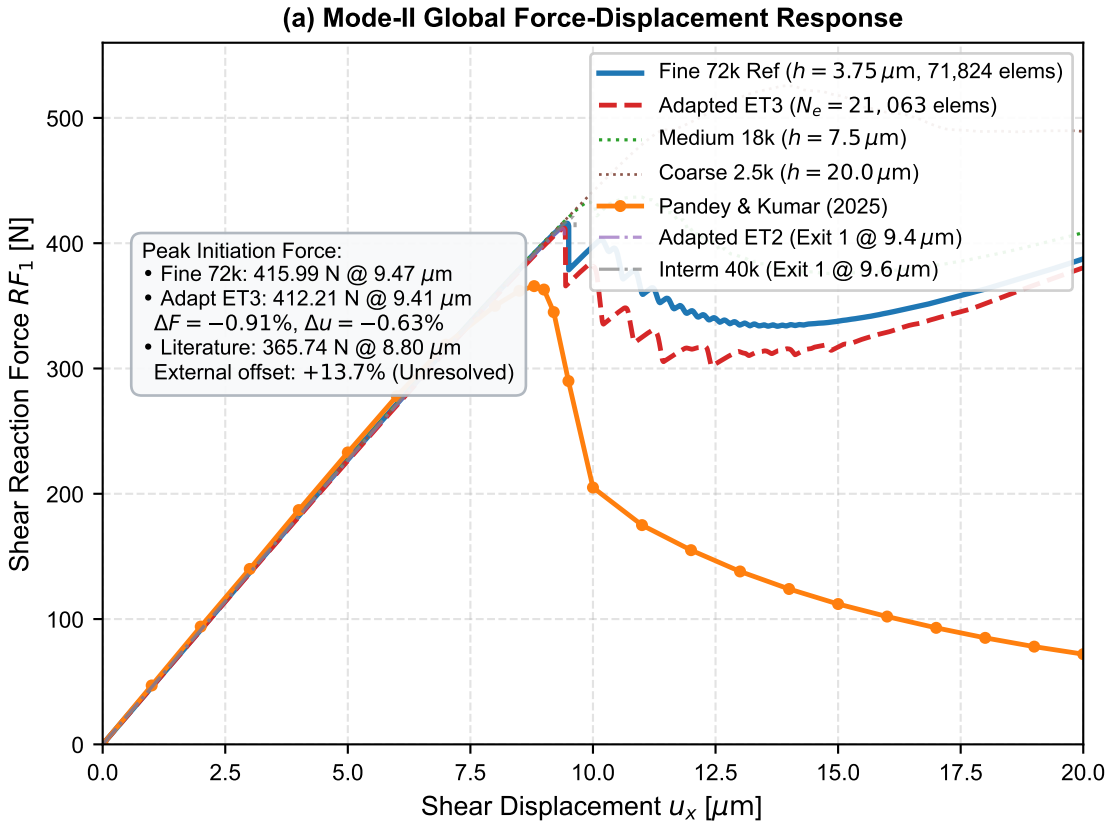
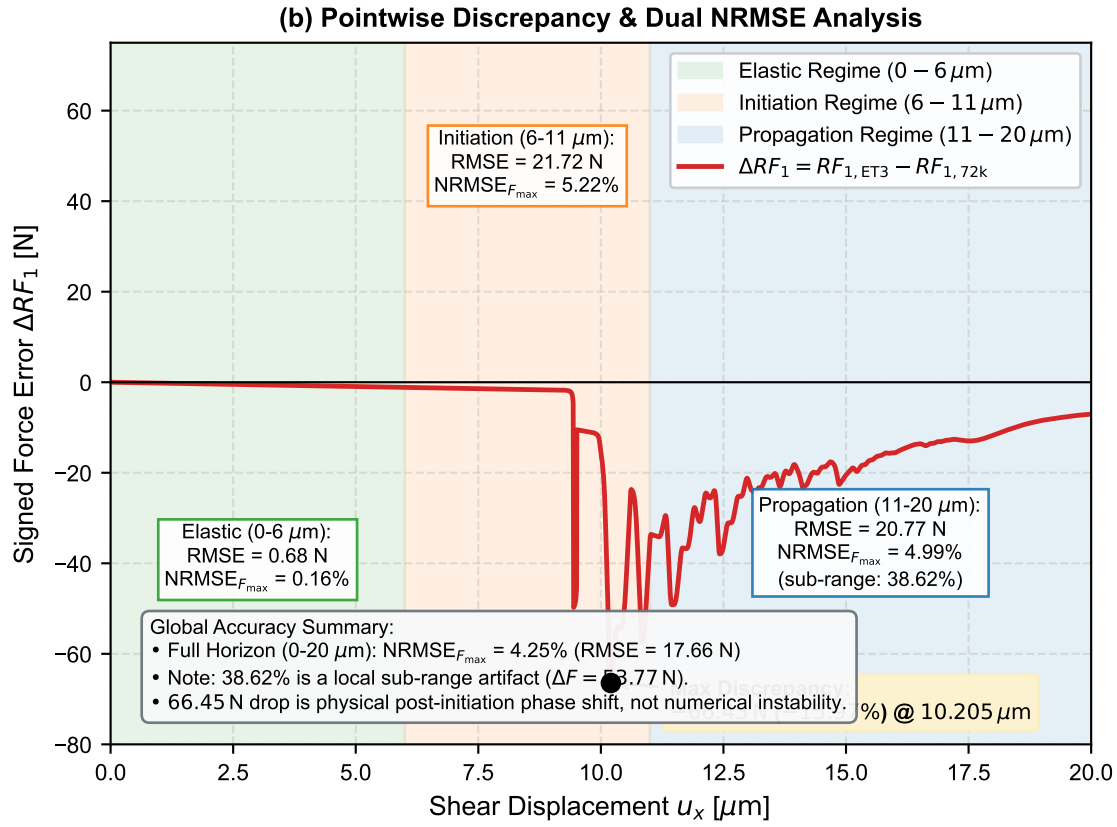
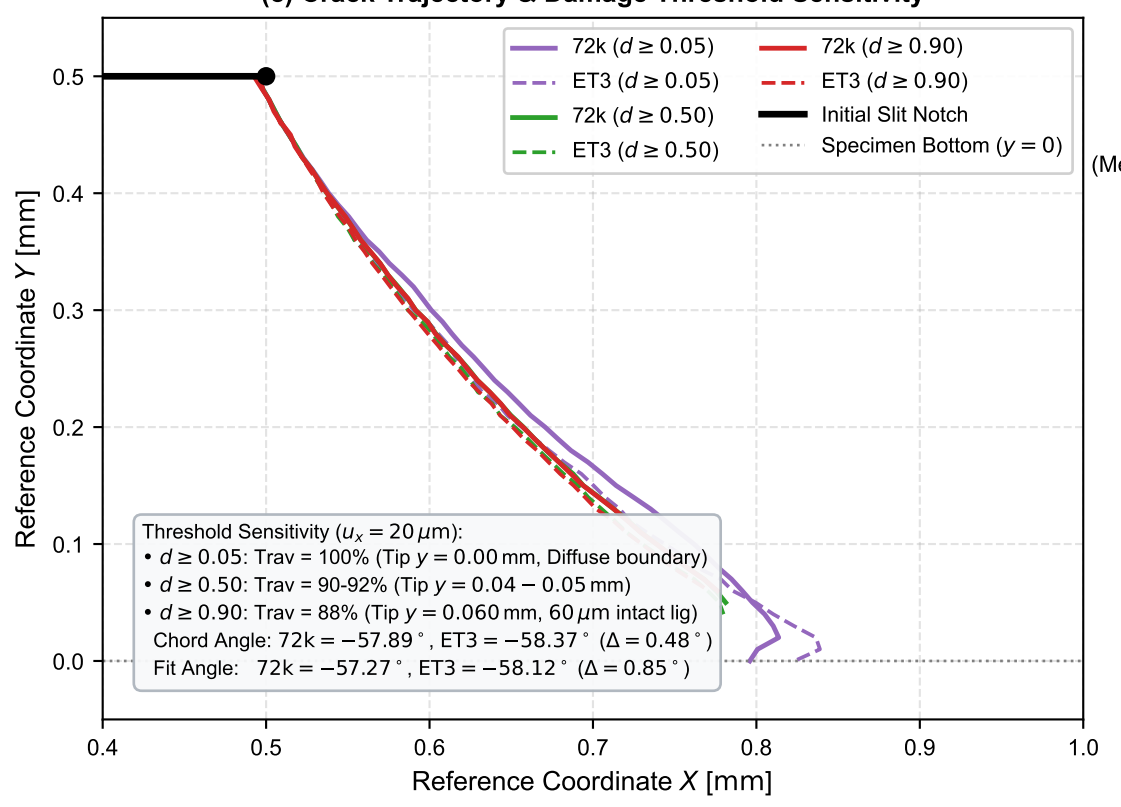
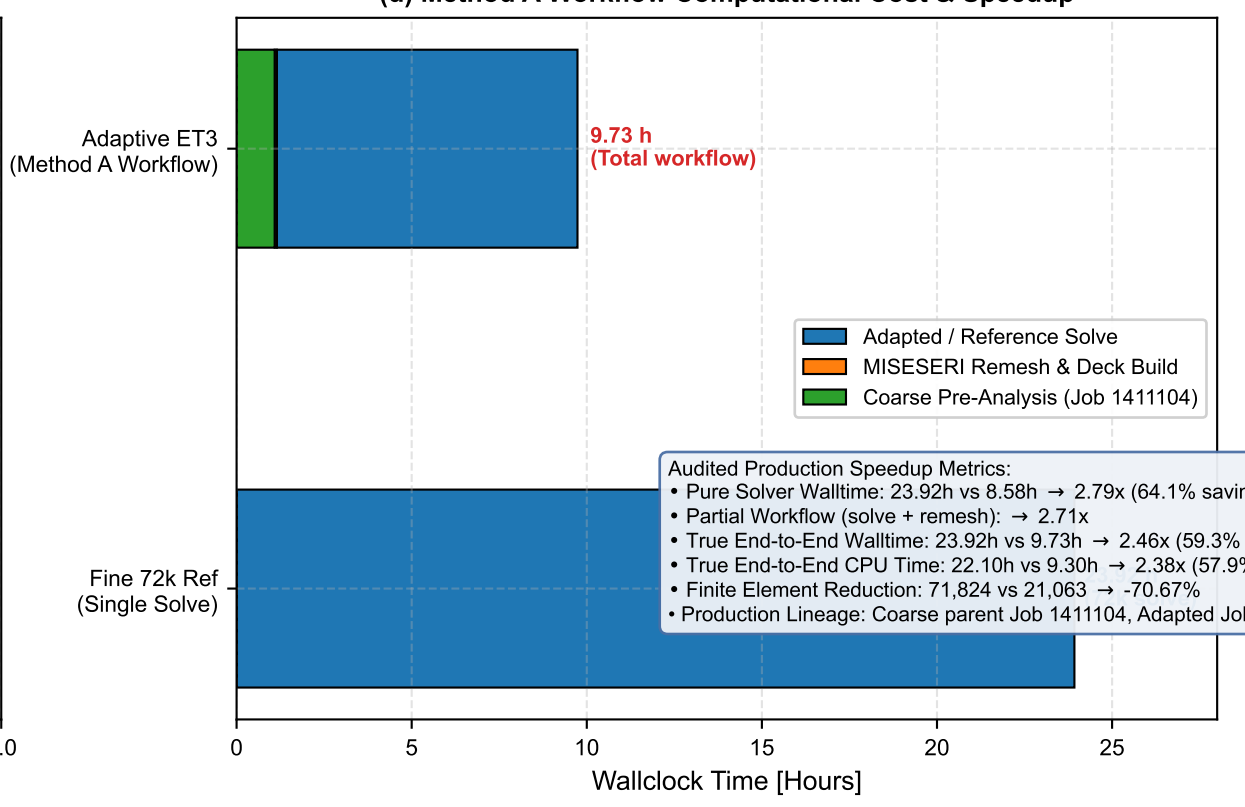

(a) Mode-II Global Force-Displacement Response
Fine 72k Ref ($h = 3.75\,\mu$m, 71,824 elems)
Adapted ET3 ($N_e = 21,063$ elems)
Medium 18k ($h = 7.5\,\mu$m)
Coarse 2.5k ($h = 20.0\,\mu$m)
Pandey & Kumar (2025)
Adapted ET2 (Exit 1 @ 9.4 μm)
Interm 40k (Exit 1 @ 9.6 μm)
Peak Initiation Force:
• Fine 72k: 415.99 N @ 9.47 μm
• Adapt ET3: 412.21 N @ 9.41 μm
$\Delta F = -0.91\%$, $\Delta u = -0.63\%$
• Literature: 365.74 N @ 8.80 μm
External offset: +13.7% (Unresolved)
Shear Reaction Force RF_1 [N]
Shear Displacement u_x [μm]


(b) Pointwise Discrepancy & Dual NRMSE Analysis
Elastic Regime (0 – 6 μm)
Initiation Regime (6 – 11 μm)
Propagation Regime (11 – 20 μm)
$\Delta RF_1 = RF_{1,\mathrm{ET3}} - RF_{1,\mathrm{72k}}$
Initiation (6-11 μm):
RMSE = 21.72 N
NRMSE$_{F_{max}}$ = 5.22%
Elastic (0-6 μm):
RMSE = 0.68 N
NRMSE$_{F_{max}}$ = 0.16%
Propagation (11-20 μm):
RMSE = 20.77 N
NRMSE$_{F_{max}}$ = 4.99%
(sub-range: 38.62%)
Global Accuracy Summary:
• Full Horizon (0-20 μm): NRMSE$_{F_{max}}$ = 4.25% (RMSE = 17.66 N)
• Note: 38.62% is a local sub-range artifact (ΔF = 58.77 N).
• 66.45 N drop is physical post-initiation phase shift, not numerical instability.
Signed Force Error ΔRF_1 [N]
Shear Displacement u_x [μm]


(c) Crack Trajectory & Damage Threshold Sensitivity
72k ($d \geq 0.05$)
ET3 ($d \geq 0.05$)
72k ($d \geq 0.50$)
ET3 ($d \geq 0.50$)
72k ($d \geq 0.90$)
ET3 ($d \geq 0.90$)
Initial Slit Notch
Specimen Bottom ($y = 0$)
Threshold Sensitivity ($u_x = 20\,\mu$m):
• $d \geq 0.05$: Trav = 100% (Tip $y = 0.00$ mm, Diffuse boundary)
• $d \geq 0.50$: Trav = 90-92% (Tip $y = 0.04 - 0.05$ mm)
• $d \geq 0.90$: Trav = 88% (Tip $y = 0.060$ mm, 60 μm intact lig)
Chord Angle: 72k = −57.89°, ET3 = −58.37° ($\Delta = 0.48°$)
Fit Angle: 72k = −57.27°, ET3 = −58.12° ($\Delta = 0.85°$)
Reference Coordinate Y [mm]
Reference Coordinate X [mm]


(d) Method A Workflow Computational Cost & Speedup
Adaptive ET3
(Method A Workflow)
9.73 h
(Total workflow)
Fine 72k Ref
(Single Solve)
Adapted / Reference Solve
MISESERI Remesh & Deck Build
Coarse Pre-Analysis (Job 1411104)
Audited Production Speedup Metrics:
• Pure Solver Walltime: 23.92h vs 8.58h → 2.79x (64.1% savings)
• Partial Workflow (solve + remesh): → 2.71x
• True End-to-End Walltime: 23.92h vs 9.73h → 2.46x (59.3% savings
• True End-to-End CPU Time: 22.10h vs 9.30h → 2.38x (57.9% saving
• Finite Element Reduction: 71,824 vs 21,063 → -70.67%
• Production Lineage: Coarse parent Job 1411104, Adapted Job 14112
Wallclock Time [Hours]